

The Hidden Housing Portfolio

A Spatial and Temporal Analysis of King County House Sales

EDA Project
Client: Amy Williams

Client Goal

Analyse how house sale prices vary across central and outskirt areas over time to support his diversified portfolio strategy.

What the client want to achieve?

Develop a diversified house-selling strategy based on:

- High-value properties: Focus on houses in the **top 10% of sale prices Central locations**
- **Outskirt properties:** Properties outside the central area, with a focus on **average-priced houses**
- **Timing** Sales distributed across **multiple periods**

Assumptions/Hypotheses

H1: Houses in central areas are more likely to belong to the top 10% of property values.

Test: Compare the proportion of top-10% houses in central vs. outskirt areas.

H2: High-value central houses are sold across multiple time periods rather than being concentrated in a single period.

Test: Assess sales over time

H3: Houses in outskirts areas are more likely to have average sale prices than houses in central areas.

Test: Compare the sale-price distributions of: Central properties and Outskirt properties

Data analysis Method (1/1)

Data Assessment

Step 1. Inspect

Available variables

- Data types
- Identifiers
- Potential missing values

Step 2. Count

Record counts were checked **before the join** to establish a baseline.

Step 3. Join

Processed in Dbeaver

Step 4. Validate

The joined record count was compared with the original sales count to identify:

Missing matches

Unexpected duplication

Multiple sales of the same property

Dataset

king_county_house_details

king_county_house_sales

Main variables used

lat, long, (area_type from distance_to_centre)

Price (high_value), **date**

Preparing the data for analysis

- Repeated property IDs retained
- Missing categorical values retained as **unknown**

Important decision

Repeated property IDs were **not removed** because they may represent:

The same property being sold multiple times at different dates.

This is particularly important for the **temporal analysis**.

Data analysis Method (1/2)

Three dimensions of the housing portfolio
Have been assessed:

LOCATION

Central and Outskirts

TIME

Multiple sales periods Analytical framework

PRICE

Top 10% high-priced houses vs. Average-priced
properties

Price - Location - Time

This allows us to display where the **expensive houses
located**, and what happens **over time** ?

Defining Central & Outskirt Areas

The geographical centre is calculated using the median
latitude and longitude of the King County house-sales
dataset.

Central area

Properties located:

≤ 13.89 km from the median geographic centre

Outskirt area

Properties located:

> 13.89 km from the median geographic centre

CENTRAL

Higher proximity to the geographic centre
and

OUTSKIRTS

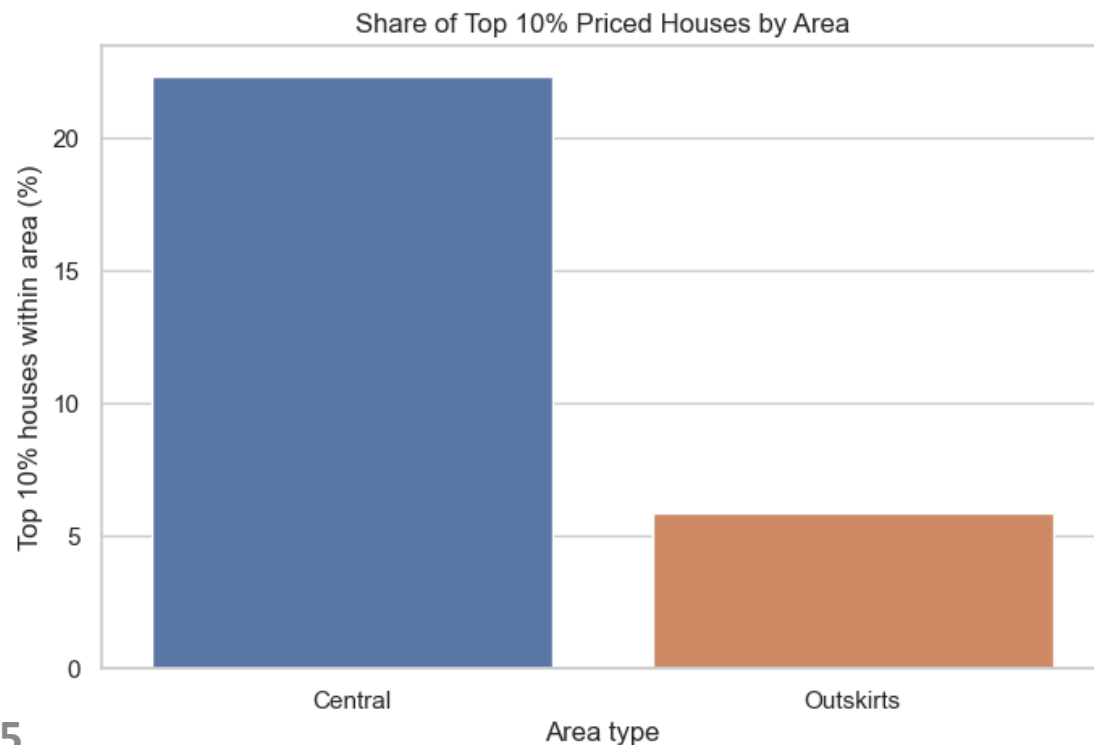
Greater distance from the centre

Key Insights (1/1)

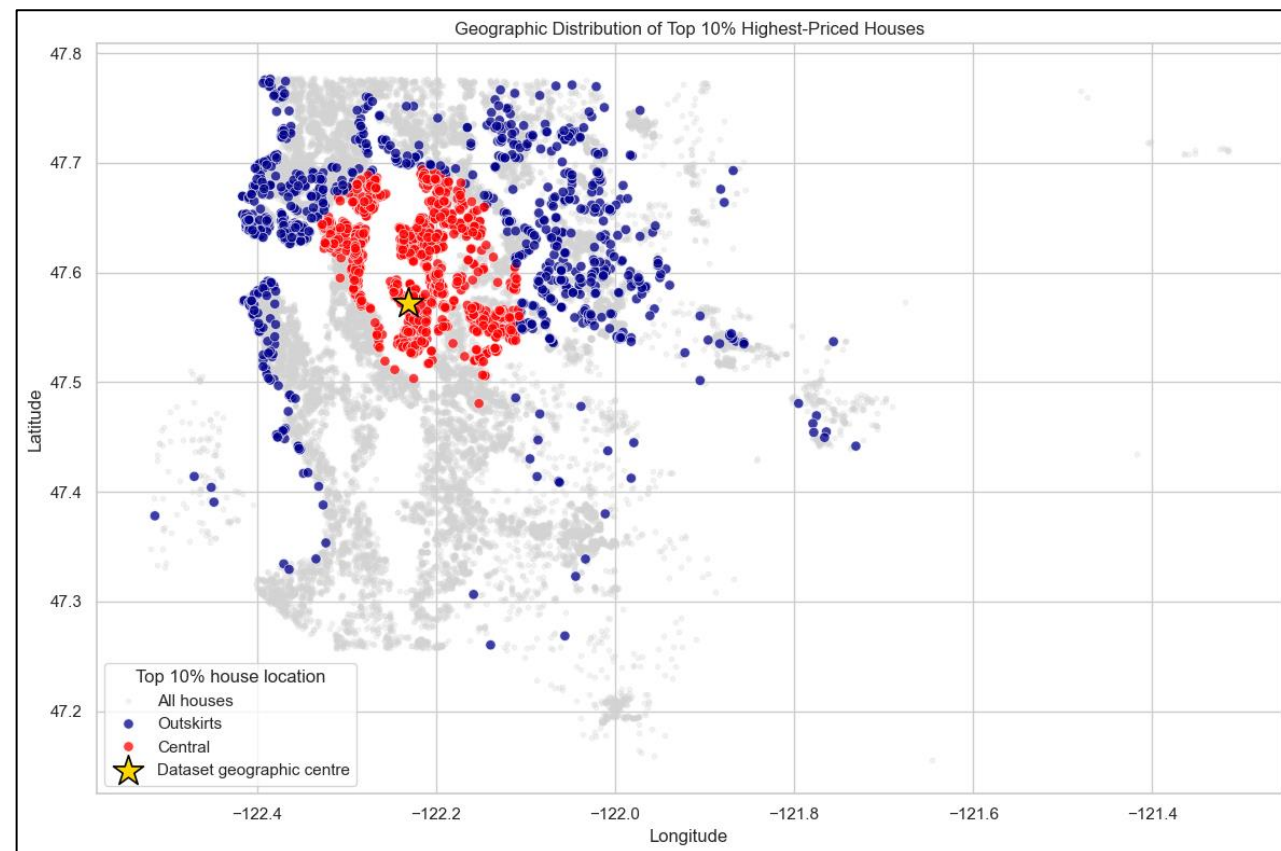
High-value central houses

H1: Central houses are more likely to be in the top 10% of prices

Top-10% properties are substantially more prevalent in central areas: 22.37% of central houses are in the top 10%, compared with only 5.88% of outskirt houses.



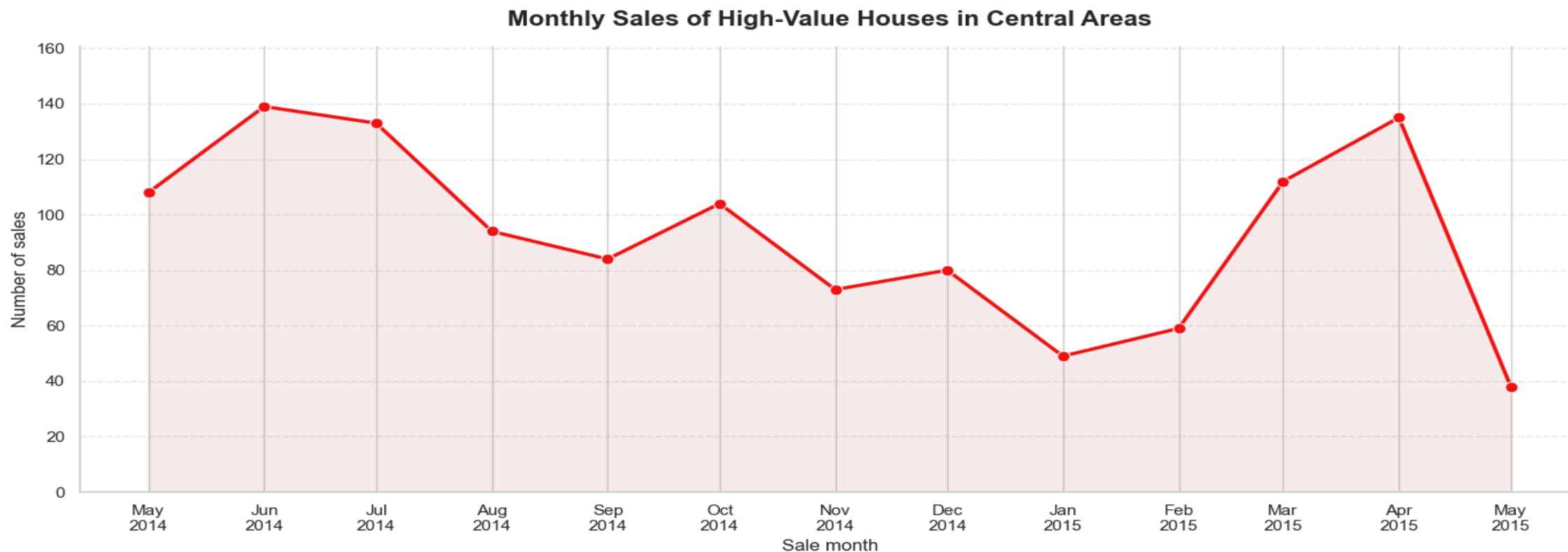
Id	Area_type	Number_of_top_10_houses	Percentage_within_area (%)
0	Central	1208	22.37
1	Outskirts	953	5.88



Key Insights (1/2)

Sales Over Time

H2: High-value central houses are sold across several periods



High-value central properties are sold across several periods, rather than being concentrated in a single period

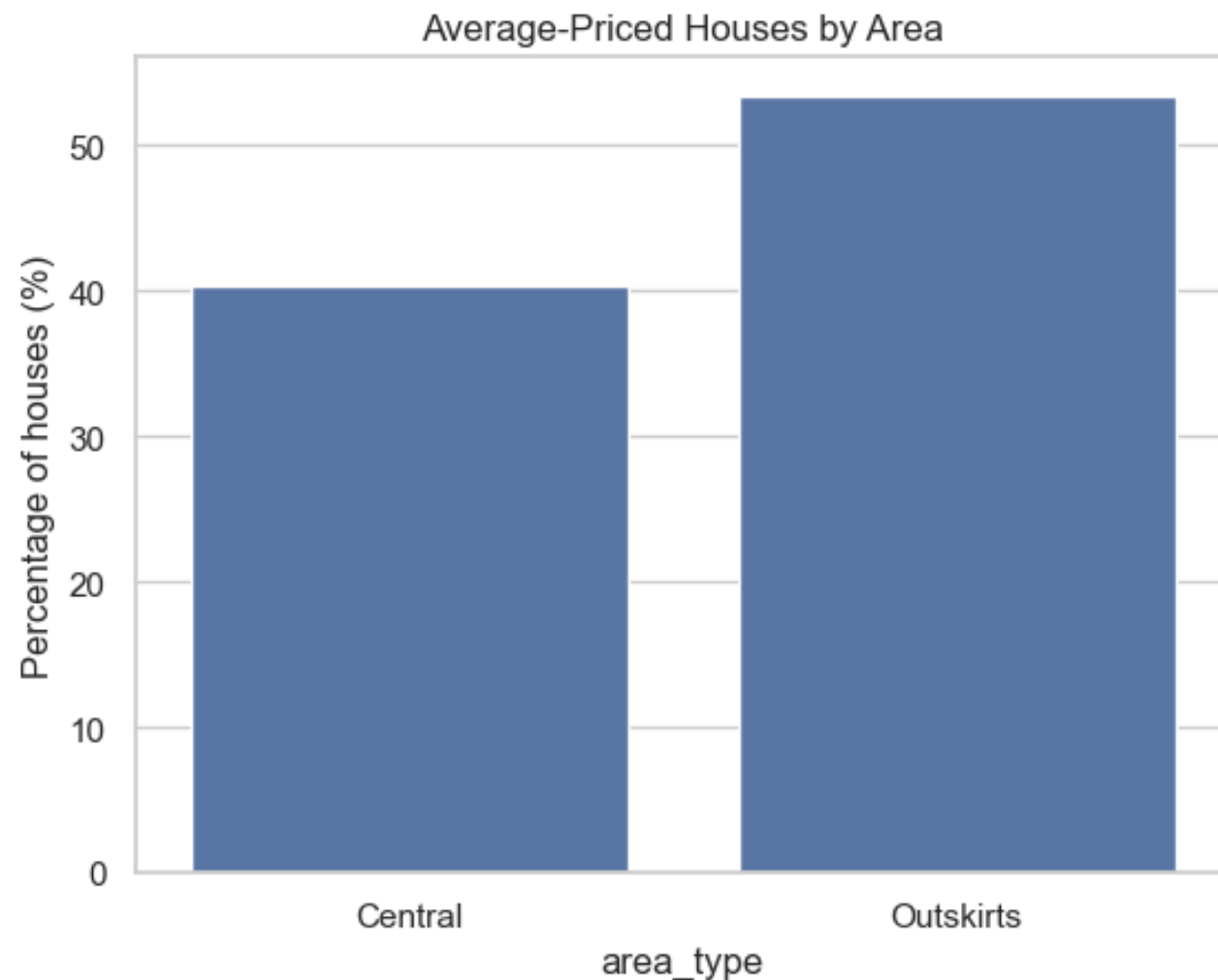
Key Insights (1/3)

Average-priced outskirts houses

H3: Outskirts houses are more likely to have average prices

id	Area type	Average price (%)
0	Central	40.37
1	Outskirts	53.42

Average-priced properties are more prevalent in the outskirts: 53.42% compared with 40.37% in central areas



Recommendations

Based on the Key Insights :

we recommend a diversified portfolio strategy **combining** high-value properties in central areas while **incorporating** average-priced properties in the outskirts, by Keep monitoring how **both segments are distributed over time.**

a. Central: High-value

Prioritize top-10% properties in central areas.

b. Outskirts: Average-priced properties

Incorporating average-priced properties in outskirts areas.

c. Across time: monitor both segments

Distribute both property segments across multiple periods.

